

Emergency kit for CTL

François Laroussinie

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1 CTL

Let AP be a finite set of atomic propositions. CTL formulas are defined as follows :

$$\text{CTL} \ni \varphi, \psi ::= p \mid \varphi \vee \psi \mid \neg \varphi \mid \mathbf{EX} \varphi \mid \mathbf{AX} \varphi \mid \mathbf{E}\varphi \mathbf{U} \psi \mid \mathbf{A}\varphi \mathbf{U} \psi$$

with $p \in \text{AP}$.

A CTL formula is interpreted over a state of a Kripke structure $\mathcal{S} = \langle S, s_0, \rightarrow, \ell \rangle$:

- $\mathcal{S}, s \models p$ iff $p \in \ell(s)$
- $\mathcal{S}, s \models \varphi \vee \psi$ iff $\mathcal{S}, s \models \varphi$ or $\mathcal{S}, s \models \psi$
- $\mathcal{S}, s \models \neg \varphi$ iff $\mathcal{S}, s \not\models \varphi$
- $\mathcal{S}, s \models \mathbf{EX} \varphi$ iff there exists $s \rightarrow s'$ s.t. $\mathcal{S}, s' \models \varphi$
- $\mathcal{S}, s \models \mathbf{E}\varphi \mathbf{U} \psi$ iff there exists some $\rho \in \text{Exec}(s)$ s.t. $\exists j \geq 0 : \mathcal{S}, \rho(j) \models \psi$ and $\forall 0 \leq k < j$, we have : $\mathcal{S}, \rho(k) \models \varphi$
- $\mathcal{S}, s \models \mathbf{A}\varphi \mathbf{U} \psi$ iff for any $\rho \in \text{Exec}(s)$, we have $\exists j \geq 0 : \mathcal{S}, \rho(j) \models \psi$ and $\forall 0 \leq k < j$, we have : $\mathcal{S}, \rho(k) \models \varphi$

where $\text{Exec}(s)$ denotes the set of infinite executions from s .

2 Model checking algorithm

Model-checking : $\mathcal{S} \models \varphi$

input : $\mathcal{S} = \langle S, s_0, \rightarrow, \ell \rangle$ a Kripke structure, $\varphi \in \text{CTL}$

output : Yes iff there exists $w \in W(\mathcal{S})$ such that $\mathcal{D}, s_0 \models \varphi$.

The model-checking algorithm for CTL consists in **marking** every state of the structure by the subformulas it satisfies. In the following we use $s.\psi$ to denote the variable storing the truth value of ψ at $s \in S$. We proceed inductively over the formula as described in Algorithms 1 and 2.

Let Φ be a CTL formula and $\mathcal{S} = \langle S, s_0, \rightarrow, \ell \rangle$ be a Kripke structure. The complexity of this algorithm is stated as follows : for Boolean cases, we get procedures in $O(|S|)$ and for temporal operators, we have procedures in $O(|S| + |\rightarrow|)$. We use $|\mathcal{S}|$ to denote $|S| + |\rightarrow|$. The overall complexity of the model-checking algorithm is in $O(|\Phi| \cdot |\mathcal{S}|)$.

The satisfiability of CTL is also decidable but the complexity is higher. Finally we have :

Theorem 1. — *CTL Model-checking is P-complete.*

— *CTL satisfiability is EXPTIME-complete.*

Procedure $\text{Mark}(\psi)$ (part 1.)

```

case  $\psi = p$  : do
  foreach  $s \in S$  do
    if  $p \in \ell(s)$  then  $s.\psi := \text{true};$ 
    else  $q.\psi := \text{false};$ 
  end
case  $\psi = \neg\psi_1$  : do
   $\text{Mark}(\psi_1);$ 
  foreach  $s \in S$  do
     $s.\psi := \neg s.\psi_1;$ 
  end
case  $\psi = \psi_1 \vee \psi_2$  : do
   $\text{Mark}(\psi_1); \text{Mark}(\psi_2);$ 
  foreach  $s \in S$  do
     $s.\psi := \neg s.\psi_1 \vee \neg s.\psi_2;$ 
  end
case  $\psi = \text{EX } \psi_1$  : do
   $\text{Mark}(\psi_1);$ 
  foreach  $s \in S$  do  $s.\psi := \text{false};$ 
  foreach  $s \rightarrow s'$  do
    if  $s'.\psi_1$  then  $s.\psi := \text{true};$ 
  end
case  $\psi = \text{AX } \psi_1$  : do
   $\text{Mark}(\psi_1);$ 
  foreach  $s \in S$  do  $s.\psi := \text{true};$ 
  foreach  $s \rightarrow s'$  do
    if  $\neg s'.\psi_1$  then  $s.\psi := \text{false};$ 
  end

```

Algorithm 1 : Model-checking for CTL (part 1)

Procedure Mark(ψ) (part 1.)

```

case  $\psi = \mathbf{E}\psi_1 \mathbf{U} \psi_2$  : do
  Mark( $\psi_1$ ); Mark( $\psi_2$ );  $L := \emptyset$ ;
  foreach  $s \in S$  do
    if  $s.\psi_2$  then  $s.\psi := \text{true}$ ;  $L := L \cup \{s\}$ ;
    else  $s.\psi := \text{false}$ ;
  end
  while  $L \neq \emptyset$  do
    pick a state  $s$  in  $L$ ;
    foreach  $s' \rightarrow s$  do
      if  $s'.\psi_1 \wedge \neg s'.\psi$  then  $L := L \cup \{s'\}$ ;  $s'.\psi := \text{true}$ ;
    end
  end
case  $\psi = \mathbf{A}\psi_1 \mathbf{U} \psi_2$  : do
  Mark( $\psi_1$ ); Mark( $\psi_2$ );  $L := \emptyset$ ;
  foreach  $s \in S$  do
     $s.\text{nb} := \text{deg}^-(s)$ ; //  $\text{deg}^-(s)$  is the out-degree of  $s$ 
    if  $s.\psi_2$  then  $s.\psi := \text{true}$ ;  $L := L \cup \{s\}$ ;
    else  $s.\psi := \text{false}$ ;
  end
  while  $L \neq \emptyset$  do
    pick a state  $s$  in  $L$ ;
    foreach  $s' \rightarrow s$  do
       $s'.\text{nb} := s'.\text{nb} - 1$ ;
      if  $s'.\psi_1 \wedge s'.\text{nb} = 0 \wedge \neg s'.\psi$  then  $L := L \cup \{s'\}$ ;  $s'.\psi := \text{true}$ ;
    end
  end

```

Algorithme 2 : Model-checking for CTL (part 2)